

Experiment 5

Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily leading to multiple integrations per day.

Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins Creating a job:

Naming the job and setting it as freestyle:

Selecting build type as “Execute shell”:

Entering a simple command for the shell execution:

Applying and saving the project configuration:

Building the project:

Console output (after building):

Example 1.2: Taking parameters through files Contents of script example1.cmd:

Executing script example1.cmd on the terminal:

Modifying the Jenkins project to execute the script while supplying required parameters:

Console output after building the modified project:

Example 2

Example 2.1: Running a Java program under Jenkins Creating a simple Java program:

Compiling and running the program on the terminal:

Creating a new freestyle project:

Configure new project:

Console output after building:

Example 3

Example 3.1: Parameterise build

Creating a new freestyle project:

Enabling parameterisation and adding a String parameter:

Configuring the string parameter as Fname:

Adding a choice parameter and configuring it as City with the following choices:

Creating a script which takes 2 arguments for name and city:

Configuring build steps:

Entering parameters for build:

Console output after building:

Example 3.2: Running a Java program with parameters Creating

a Java program with an input argument:

Testing the program on the terminal:

Creating a new freestyle project:

Parameterise the project by adding a string parameter as follows:

Configure the build steps:

Entering the parameter for the build:

Console output after building:

Example 5

Example 5.1: Running a Python program Creating

a simple Python script:

Running the Python script on the terminal:

Creating a new freestyle project:

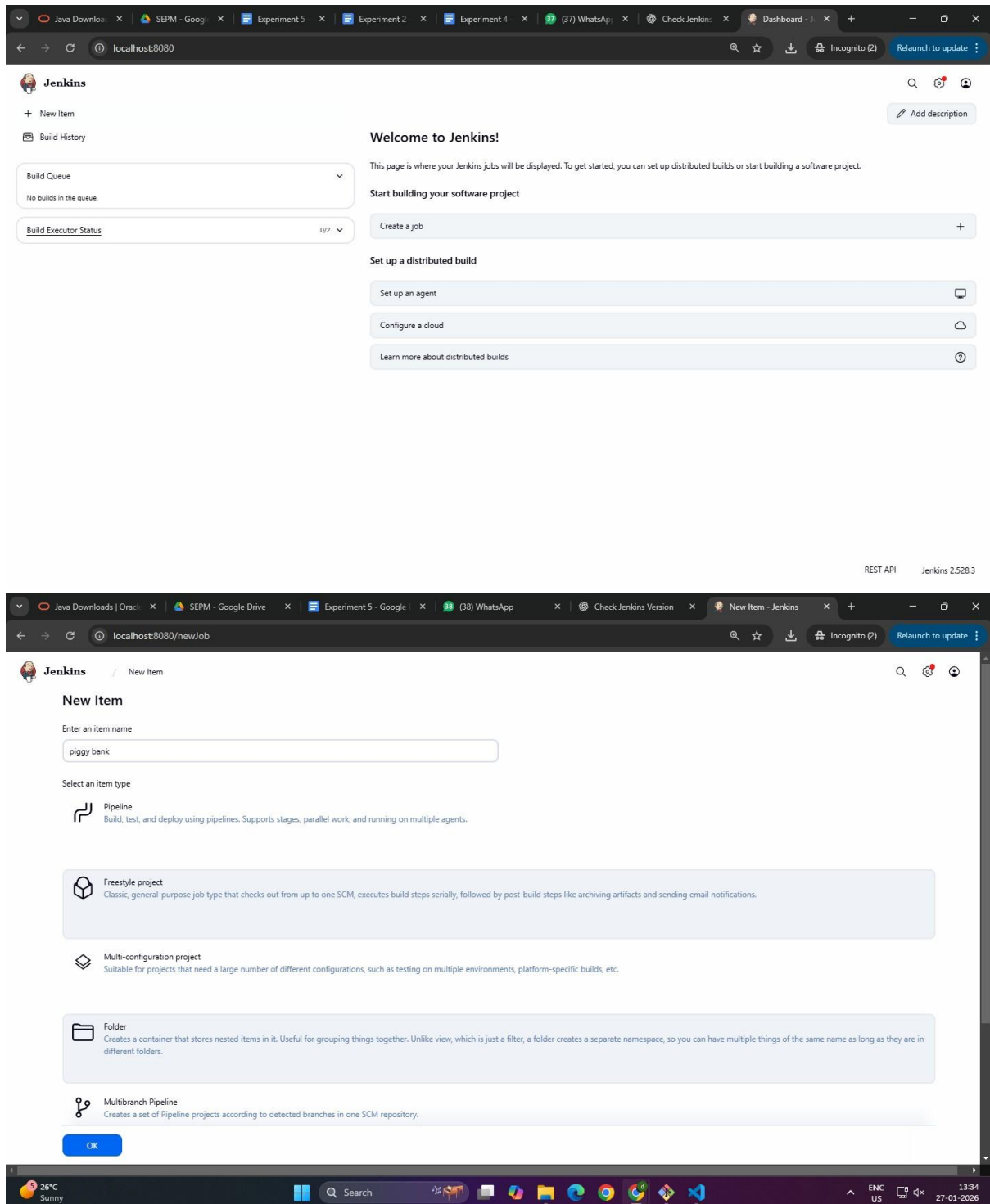
Parameterising the project with a string parameter as follows:

Configuring the build steps:

Setting the parameter for the build:

Console output after building:

Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.



The image shows two screenshots of the Jenkins web interface. The top screenshot is the Jenkins Dashboard at localhost:8080. It features a sidebar with 'New Item' and 'Build History' links. The main area displays 'Welcome to Jenkins!' and instructions on how to start building software projects. It includes buttons for 'Create a job' and 'Set up a distributed build' (with sub-options like 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'). The bottom screenshot shows the 'New Item' page at localhost:8080/newJob. It prompts the user to 'Enter an item name' (with 'piggy bank' entered) and 'Select an item type'. The available types are Pipeline, Freestyle project, Multi-configuration project, Folder, and Multibranch Pipeline, each with a brief description. An 'OK' button is at the bottom.

Jenkins Dashboard (localhost:8080)

REST API Jenkins 2.528.3

New Item

Enter an item name

piggy bank

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.

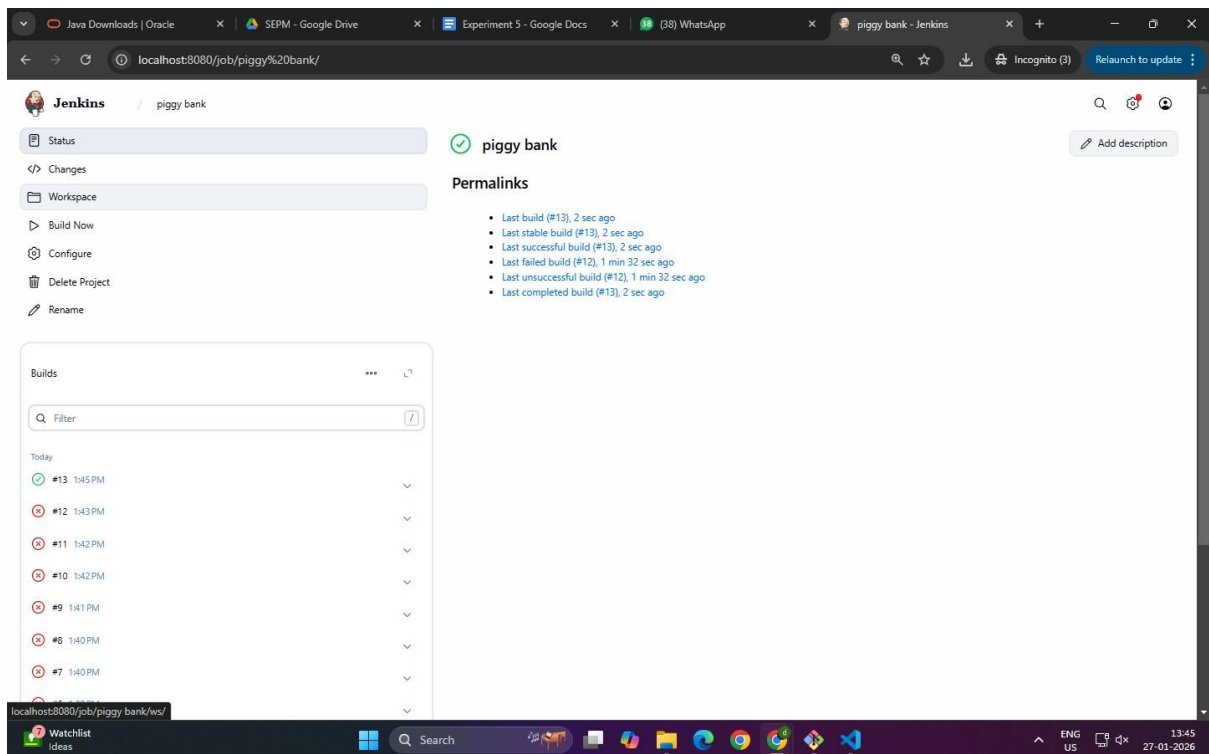
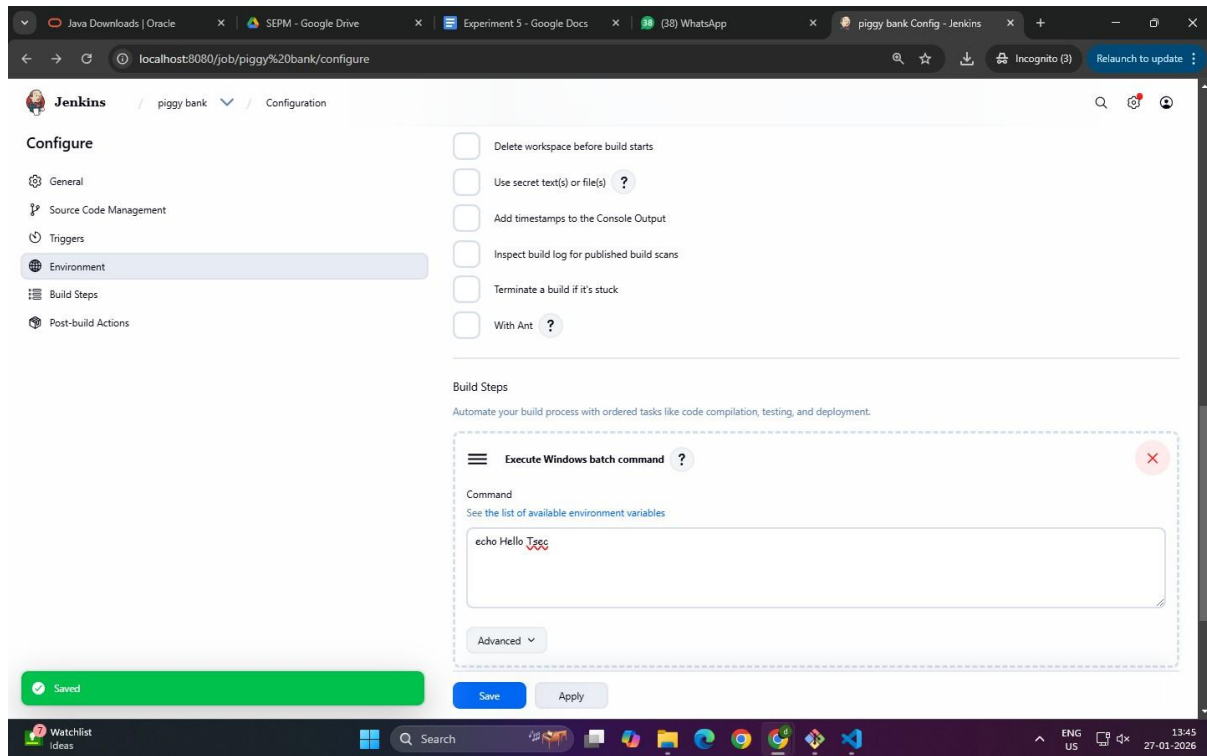
OK

The image displays two screenshots of the Jenkins web interface, showing the configuration page for a job named "piggy bank".

Top Screenshot: The "General" tab is selected in the left sidebar. The "General" section is titled "General" and has a status of "Enabled". The "Description" field is empty. Below it, there are several checkboxes: "Discard old builds", "GitHub project", "This project is parameterized", "Throttle builds", and "Execute concurrent builds if necessary". An "Advanced" dropdown menu is visible. The "Source Code Management" section is also visible, with "None" selected. The "Save" and "Apply" buttons are at the bottom.

Bottom Screenshot: The "Build Steps" tab is selected in the left sidebar. The "Command" section is titled "Command" and has a status of "Enabled". The "Command" field contains the text "echo 'Hello Tsec'". Below it, there is an "Advanced" dropdown menu and a "+ Add build step" button. The "Post-build Actions" section is also visible, with a "+ Add post-build action" button. The "Save" and "Apply" buttons are at the bottom.

A green "Saved" notification bar is visible at the bottom of the bottom screenshot. The bottom status bar shows "REST API" and "Jenkins 2.528.3".



The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL `localhost:8080/job/piggy%20bank/lastBuild/console`. The Jenkins page title is "piggy bank #13 Console - Jenkins". The left sidebar contains links for "Status", "Changes", "Console Output" (selected), "Edit Build Information", "Delete build #13", "Timings", and "Previous Build". The main content area is titled "Console Output" with a green checkmark icon. It includes buttons for "Download", "Copy", and "View as plain text". The console output text is as follows:

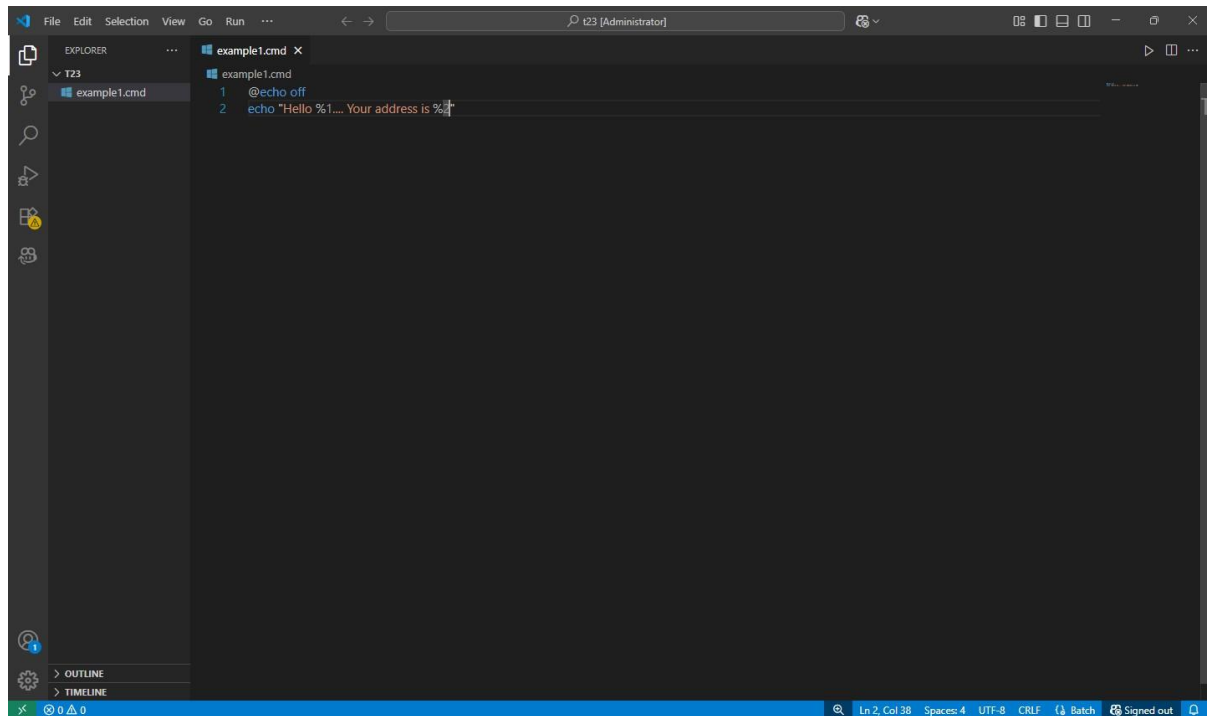
```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\piggy bank
[piggy bank] $ cmd /c call C:\WINDOWS\TEMP\Jenkins8989928918577088258.bat

C:\ProgramData\Jenkins\jenkins\workspace\piggy bank>echo Hello Tsec
Hello Tsec

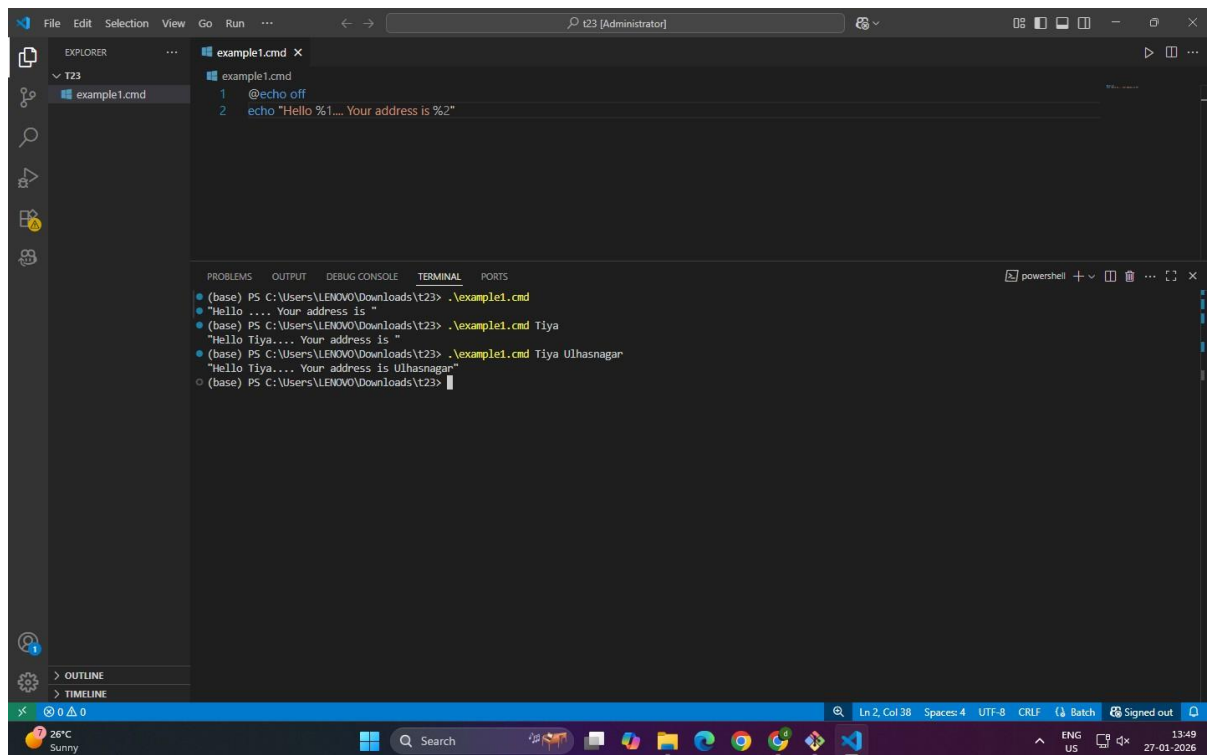
C:\ProgramData\Jenkins\jenkins\workspace\piggy bank>exit 0
Finished: SUCCESS
```

At the bottom right of the Jenkins page, it says "REST API" and "Jenkins 2.528.3". The Windows taskbar at the bottom shows the Start button, a search bar, and several application icons. The system tray on the right shows the language as "ENG US", the date as "27-01-2026", and the time as "13:45".

1.2



```
File Edit Selection View Go Run ... t23 [Administrator]
EXPLORER example1.cmd X
  T23
    example1.cmd
example1.cmd
1 @echo off
2 echo "Hello %1.... Your address is %2"
```



```
File Edit Selection View Go Run ... t23 [Administrator]
EXPLORER example1.cmd X
  T23
    example1.cmd
example1.cmd
1 @echo off
2 echo "Hello %1.... Your address is %2"

TERMINAL
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd
"Hello .... Your address is "
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya
"Hello Tiya.... Your address is "
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya Ulhasnagar
"Hello Tiya.... Your address is Ulhasnagar"
(base) PS C:\Users\LENOVO\Downloads\t23>
```

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Batch: T23

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Example 2

The image shows two screenshots of the Jenkins web interface. The top screenshot displays the 'Configure' page for a job named 'Example 2'. The left sidebar shows the configuration menu with 'Environment' selected. The main area has several checkboxes for build options, all of which are currently unchecked. Below these is the 'Build Steps' section, which contains a single step named 'Execute Windows batch command'. The command field for this step contains the text 'C:\Users\LENOVO\Downloads\T23\Example2.java'. The bottom screenshot shows the 'Example 2' job page. The left sidebar has 'Status' selected. The main area shows the job's name 'Example 2' and a 'Permalinks' section. Below this is a 'Builds' section showing a single build with a status of '1' and a time of '2:00 PM'. At the bottom of the page, a green notification bar states 'Build scheduled'. The bottom status bar shows the REST API endpoint 'localhost:8080/job/Example 2/build?delay=0sec' and the Jenkins version 'Jenkins 2.528.3'.

Jenkins Configuration Page (Example 2)

Configure

- General
- Source Code Management
- Triggers
- Environment**
- Build Steps
- Post-build Actions

Build Options:

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Terminate a build if it's stuck
- ☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

C:\Users\LENOVO\Downloads\T23\Example2.java

Advanced

Save Apply

Jenkins Job Page (Example 2)

Example 2

Permalinks

Builds

Today

1 2:00 PM

Build scheduled

localhost:8080/job/Example 2/build?delay=0sec

REST API Jenkins 2.528.3

Example 3

The image displays two screenshots of the Jenkins web interface, showing the process of creating and configuring a new job.

Top Screenshot: New Item Page

- The browser address bar shows `localhost:8080/view/all/newJob`.
- The page title is "New Item".
- Under "Enter an item name", the text "Example 3" is entered.
- Under "Select an item type", the "Freestyle project" is selected. Its description reads: "Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications."
- Other visible options include "Pipeline", "Multi-configuration project", "Folder", and "Multibranch Pipeline".
- An "OK" button is at the bottom.

Bottom Screenshot: Configuration Page

- The browser address bar shows `localhost:8080/job/Example%203/configure`.
- The page title is "Configure".
- The left sidebar shows the configuration tree with "General" selected.
- Under "This project is parameterized", the checkbox is checked. A dropdown menu is open, showing options: "Boolean Parameter", "Choice Parameter", "Credentials Parameter", "File Parameter", "Multi-line String Parameter", "Password Parameter", "Run Parameter", and "String Parameter".
- Other visible options include "GitHub project", "Triggers", and "Build after other projects are built".
- "Save" and "Apply" buttons are at the bottom.

The image displays two screenshots of the Jenkins web interface, specifically the configuration page for a job named 'Example 3'. The browser's address bar shows the URL 'localhost:8080/job/Example%203/configure'. The left sidebar contains a 'Configure' menu with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main content area shows the 'General' tab selected. At the top, a checkbox 'This project is parameterized' is checked. Below this, there are two parameter configuration sections. The first is a 'String Parameter' with the name 'fname', an empty 'Default Value' field, and an empty 'Description' field. The second is a 'Choice Parameter' with the name 'City', a list of choices including 'Bandra', 'Wileasnagar', 'Kalyan', and 'Mulund', and an empty 'Description' field. At the bottom of the configuration area are 'Save' and 'Apply' buttons. The Windows taskbar at the bottom shows the system clock as 14:04 on 27-01-2026, and the weather as 26°C Sunny.

The image shows a Jenkins configuration page for a job named 'Example 3' and a Visual Studio Code editor window.

Jenkins Configuration:

- The 'Configure' page for 'Example 3' is shown, with the 'General' tab selected.
- A checkbox 'This project is parameterized' is checked.
- Two parameters are configured:
 - String Parameter:** Name 'fname', Default Value (empty), Description (empty). Options: Plain text, Preview. Trim the string: unchecked.
 - Choice Parameter:** Name (empty).
- A green 'Saved' notification is visible.
- Buttons for 'Save' and 'Apply' are present.

Visual Studio Code Editor:

- The Explorer view shows a project structure for 't23' with files: example1.cmd, Example2.class, Example2.java, example3.cmd, and Main.class.
- The editor window shows 'example3.cmd' with the following content:

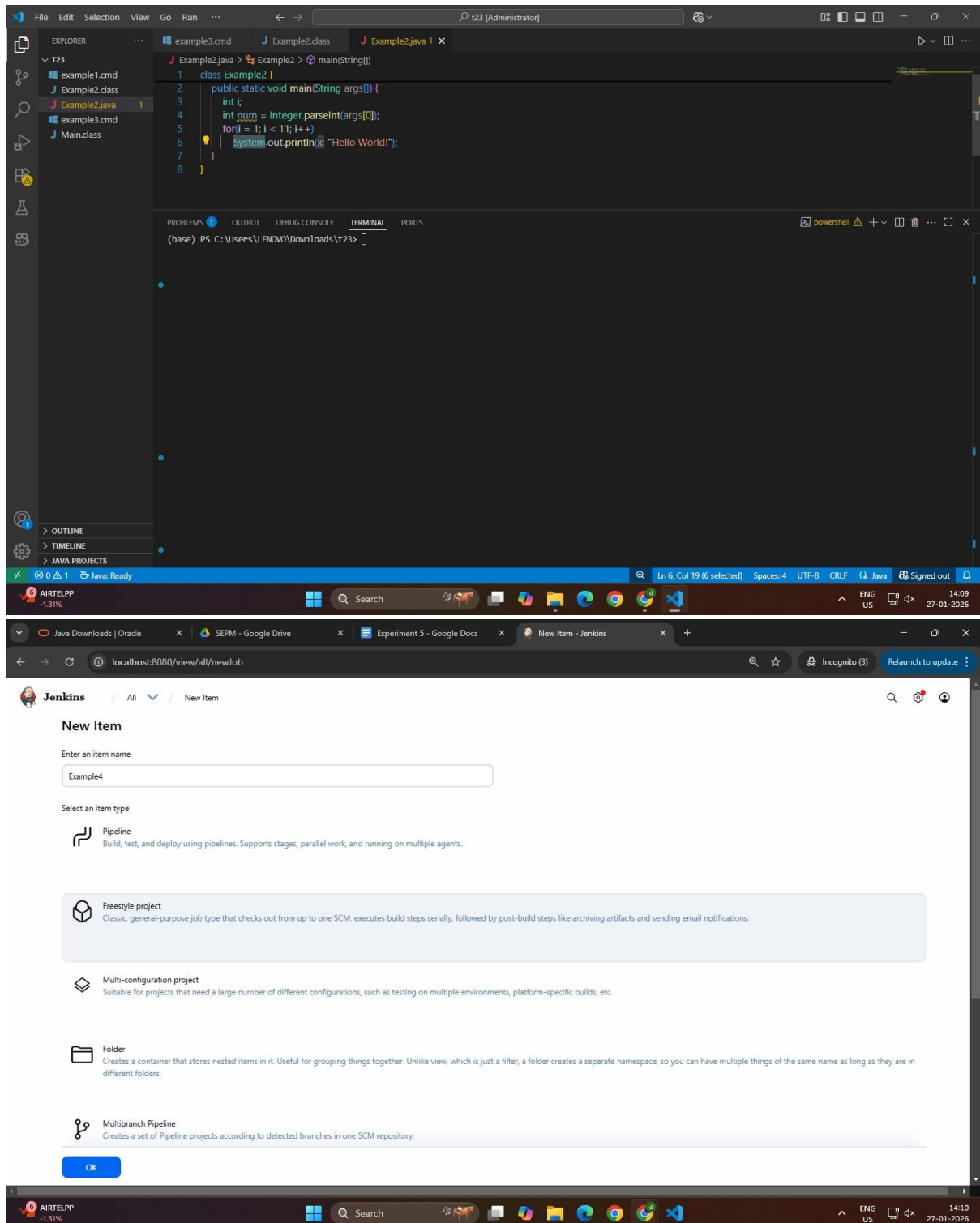
```
1 @echo off
2 echo "Hello %1.... Your city is %2"
```
- The TERMINAL view shows a PowerShell prompt: (base) PS C:\Users\LENOVO\Downloads\t23>.
- The status bar at the bottom indicates 'Ln 2, Col 29, Spaces 4, UTF-8, CRLF, Batch, Signed out'.

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3.2



The image displays two screenshots of the Jenkins web interface, showing the configuration of a job named 'Example4'.

Top Screenshot: String Parameter Configuration

- The browser address bar shows `localhost:8080/job/Example4/configure`.
- The left sidebar shows the 'Configure' page with tabs for General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions.
- The main content area shows the 'String Parameter' configuration for a parameter named 'num'. The 'Name' field is 'num', and the 'Default Value' field is empty. The 'Description' field is also empty.
- There is a checkbox for 'Trim the string' which is currently unchecked.
- Buttons for 'Save' and 'Apply' are visible at the bottom.

Bottom Screenshot: Build Steps Configuration

- The browser address bar shows `localhost:8080/job/Example4/configure`.
- The left sidebar shows the 'Configure' page with tabs for General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions.
- The main content area shows the 'Build Steps' configuration. The 'Command' field contains the text `C:\Users\LENOVO\Downloads\t23\Example4.java %num%`.
- Buttons for 'Save' and 'Apply' are visible at the bottom.
- A green status bar at the bottom left indicates 'Saved'.